

Fredrik Svensson

C++ Software Developer | Rendering Specialist



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SUMMARY

Master of Science in Engineering with a focus on C++ software development, 3D graphics, and real-time simulation. Strong academic background in game engine development, VR systems, and performance-oriented programming. Hands-on experience with Unreal Engine, DirectX rendering fundamentals, and shader development through coursework and project work.

Core Competencies:

- C++ development with a focus on real-time and performance-oriented applications.
- Understanding of 3D graphics fundamentals, rendering pipelines, and DirectX concepts.
- Experience working with Unreal Engine for VR development and interactive simulations.
- Academic and project-based exposure to shader programming and engine architecture concepts.
- Familiar with Windows development environments and Visual Studio workflows.

EDUCATION

Master of Science in Engineering: Game and Software Engineering

Blekinge Institute of Technology 2019 - Sep 2026

Karlskrona

Focus on C++ software engineering, 3D systems, and performance optimization. Master Thesis at HINTS: Investigated material properties and spatial audio in virtual environments.

Master of Science in Engineering: Game and Software Engineering (earlier version)

Blekinge Institute of Technology 2016 - 2017

Karlskrona

One year of an earlier version of the same program.

Upper secondary technical degree

Ehrens värdska Gymnasiet 2013 - 2016

Karlskrona

SKILLS

- C / C++ / C#
- DirectX / 3D Mathematics
- Visual Studio / Windows
- Python
- LUA
- ARM/Intel Assembler
- JavaScript
- Linux
- SQL
- Machine learning (Basic)
- AI (Basic)
- Github

LANGUAGES

- Swedish (Native)
- English (Fluent)

PROFESSIONAL EXPERIENCE

Teacher assistant

Blekinge Institute of Technology 2026 - 2026

Karlskrona

Teacher assistant, Course: Extended Reality (XR) Applications and Interaction Techniques.

- Facilitated lab sessions and mentored students in VR project development using Unity and Unreal Engine.

Researcher/Thesis student

Blekinge Institute of Technology 2025 - 2025

Karlskrona

Intermittent employment – Master Thesis work.

- Conducted research in real-time simulation, resulting in a peer-reviewed conference paper presented at GRIVAPP 2026.

Production line worker - Consultant

NKT, Conductor manufacturing 2020 - 2023

Karlskrona

Fixed-term and summer employment via SKILL.

- Managed production line operations, including material handling, wire/tape loading, and quality supervision.

Production line worker - Consultant

NKT - The Spinning/Cutting unit 2018 - 2018

Karlskrona

Summer employment as a consultant via Adecco.

- Operated and maintained precision cutting machinery for cable insulation production.

Summer Employee

Landstinget Blekinge Avd. 49 2016 - 2016

Karlskrona

Summer employment.

- Provided support for simpler tasks within the ward and kitchen environments.

CERTIFICATIONS

Conference Speaker

21st International Conference on Computer Graphics, Interaction and Visualization Theory and Applications (GRIVAPP 2026)

Mar 2026

PUBLICATIONS

Perceptual Evaluation of Physically-Based Sound Propagation in Virtual Environments

scitepress.org

Mar 2026

<https://www.scitepress.org/Link.aspx?doi=10.5220/0014327100004728>

Abstract: The development of spatial audio in virtual environments (VEs) has made significant progress, particularly with the introduction of physics-based sound propagation that incorporates material properties. However, research remains limited on how users perceive these audio enhancements in virtual reality (VR), especially regarding immersion. This paper investigates how material properties influence the user experience of spatial audio in VR, focusing on perceived realism, presence, and preference. A VR application was developed in Unreal Engine 5.5 using the Steam Audio plugin to simulate sound propagation across four common indoor materials: concrete, glass, carpet, and plaster. Each material was tested in three acoustic configurations. A fifth environment combining all materials was also tested. Participants freely explored the virtual space and rated each material configuration on audio realism on a 0–100 Likert scale. Three questionnaires were administered: pre-experiment, post-experiment, and the Simulator Sickness Questionnaire (SSQ). Results showed that while participants could distinguish between materials, they struggled to assess acoustic configuration differences. Sound propagation was generally perceived as natural and realistic, with materials rated between 'Fair' and 'Good'. Findings suggest that physics-based sound propagation enhances immersion in VR, though user familiarity with materials may significantly influence perception.

VOLUNTEER WORK

Vice Treasurer / Treasurer

Mammas Källare, SVEROK-affiliated association

2018 - 2021

Karlskrona

Responsible for financial administration and accounting. Vice Treasurer during 2018 and then Treasurer during 2019-2021

Nomination Committee Member

Blekinge Student Union

2017 - 2018

Karlskrona

Prepared and managed elections for the student union.

REFERENCES

References, certificates, and academic transcripts are available upon request and can be provided during the interview process.